Global attraction of the positive equilibrium in an irreversible reaction–diffusion system

Kaiwen Jin*

15 October 2026

Abstract

An irreversible reaction–diffusion system can have both a positive equilibrium and a stationary state at which the catalyst is absent. For the three-species system studied by Nguyen and Tang, we prove that every strictly positive compatible initial triple converges uniformly to the positive equilibrium whenever the conserved masses satisfy $M_2 \le M_1 < 2M_2$. The result holds on bounded connected smooth domains in every dimension and for arbitrary positive diffusion coefficients. Quadratic dissipation first removes the spatial oscillations of two species. The total catalyst mass then satisfies a logistic equation with a relative error tending to zero, which identifies the positive limit and closes the strictly positive form of the global-attraction question.

1 Introduction

We consider the reaction–diffusion system

$$\begin{aligned} a_t - d_1\Delta a &= b(c-a), \\ b_t - d_2\Delta b &= b(c-a), \qquad \text{in } \Omega, \\ c_t - d_3\Delta c &= -b(c-a) \end{aligned} \tag{1}$$

with homogeneous Neumann boundary conditions. Here $\Omega \subset \mathbb{R}^n$ is bounded, connected and smooth, $|\Omega| = 1$, and $d_1, d_2, d_3 > 0$. The concentrations a, b, c represent the irreversible chemical reaction studied in [1], in which the presence of b enables the reaction between a and c. Diffusion and the reaction preserve

$$\int_\Omega (a(t) + c(t))\,dx = M_1, \qquad \int_\Omega (b(t) + c(t))\,dx = M_2, \tag{2}$$

where $M_1 = \int_\Omega (a_0 + c_0)\,dx$ and $M_2 = \int_\Omega (b_0 + c_0)\,dx$. For nonnegative bounded data the global classical solution is uniformly bounded; this is established in [1, Theorem 2.1]. The remaining issue here is its long-time selection of equilibrium.

When $M_2 \le M_1 < 2M_2$, the conserved masses admit both the positive equilibrium

$$\left(\frac{M_1}{2},\ M_2 - \frac{M_1}{2},\ \frac{M_1}{2}\right)$$

and the boundary equilibrium $(M_1 - M_2, 0, M_2)$. Nguyen and Tang prove local exponential stability of the positive equilibrium in this regime [1, Theorem 2.7], and formulate global attraction as a conjecture after their Table 1. Their global positive-convergence result in the different regime $M_1 < M_2$ uses the mass lower bound $\int_\Omega b \ge M_2 - M_1 > 0$ [1, Theorem 2.4].

*yui.ui.siki@gmail.com

That bound is unavailable in the coexistence regime, so it does not settle whether every positive initial state selects the positive equilibrium.

We prove this selection for strictly positive initial concentrations. The fixed version of AIM Problem 416 assumes the same strictly positive $C^2(\overline{\Omega})$ data, Neumann compatibility, and unique global bounded classical solution used here. This restriction excludes the stationary boundary equilibrium: the printed nonnegative-data wording of the source conjecture permits $b_0 = 0$, for which that equilibrium cannot evolve to a positive state.

Theorem 1.1. *Let $n \geq 1$, let $\Omega \subset \mathbb{R}^n$ be a bounded connected smooth domain of volume one, and let $d_1, d_2, d_3 > 0$. Assume $a_0, b_0, c_0 \in C^2(\overline{\Omega})$ are strictly positive and satisfy the homogeneous Neumann compatibility conditions. Let (a, b, c) be the unique global bounded classical solution of (1). If $M_2 \leq M_1 < 2M_2$, then*

$$\lim_{t\to\infty} \left[\left\| a(t) - \frac{M_1}{2} \right\|_\infty + \left\| b(t) - \left(M_2 - \frac{M_1}{2} \right) \right\|_\infty + \left\| c(t) - \frac{M_1}{2} \right\|_\infty \right] = 0.$$

The masses are those in (2), and all uniform norms are over $\overline{\Omega}$, equivalently essential uniform norms over Ω. The diffusion coefficients may be unequal, and no smallness of the initial deviation from equilibrium is required. The statement is convergence for each admitted solution, with no common rate over all data.

The quadratic energy of a, c, already used in the proof of [1, Theorem 2.7], dissipates their spatial gradients even when the catalyst is small. Compactness of bounded forced heat orbits turns this dissipation into uniform spatial homogenization. The conserved masses then reduce the catalyst's total mass to a positive logistic comparison with an error controlled by those spatial oscillations. Its limit identifies the means of a, c; a final heat estimate removes the remaining spatial variation of b. Section 2 gives the two heat lemmas, and Section 3 proves the attraction theorem. Throughout, $\bar{f} = \int_\Omega f\, dx$ is the spatial average.

2 Two heat equation lemmas

For a continuous real function f on $\overline{\Omega}$, write

$$\bar{f} = \int_\Omega f\, dx, \qquad Qf = f - \bar{f}, \qquad X = C(\overline{\Omega}).$$

The volume normalization makes $\bar{f}$ the spatial average. We use the pointwise uniform norm on X and the usual $L^2(\Omega)$ norm. The Neumann heat semigroup for $u_t = d\Delta u$ is denoted by $S_d(t)$, where $d > 0$. Its constant mode is preserved, so Q commutes with $S_d(t)$.

We recall the precise heat properties used below. The Neumann realization of $-\Delta$ has a complete real orthonormal eigenbasis in $L^2(\Omega)$ and a smooth positive-time kernel; see [2, Chapter 4, Section 8, Proposition 4.56 and Theorem 4.58]. Green's identity [2, Proposition 4.49] shows that a zero eigenfunction has zero gradient. Connectedness therefore leaves only the constants in the zero eigenspace. We index the eigenvalues by

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots, \qquad \psi_0 = 1.$$

The multiplier of $S_d(t)$ on ψ_j is $e^{-d\lambda_j t}$. For C^2 functions g with $\partial_\nu g = 0$, Green's identity gives

$$\|\nabla g\|_2^2 = \sum_{j\geq 1} \lambda_j |\langle g, \psi_j \rangle|^2 \geq \lambda_1 \|Qg\|_2^2. \tag{3}$$

The coefficient inner product converges because $g, \Delta g \in L^2$.

The heat flow preserves nonnegativity. One may test its equation by a smooth convex approximation of the squared negative part, then use strong L^2 continuity at time zero. Its

continuous kernel $k_d(t,x,y)$ is consequently nonnegative: a negative kernel value would give a negative heat value for a nonnegative test function supported nearby. The constant mode gives

$$\int_\Omega k_d(t,x,y)\,dy = 1, \qquad \|S_d(t)f\|_\infty \le \|f\|_\infty.$$

At each fixed $\tau > 0$, continuity of the kernel on the compact closure product also gives

$$\|S_d(\tau)f\|_\infty \le H_{d,\tau}\|f\|_2, \qquad H_{d,\tau} := \sup_{x\in\overline{\Omega}} \left(\int_\Omega k_d(\tau,x,y)^2\,dy\right)^{1/2} < \infty. \tag{4}$$

Bounded input sets in X have uniformly bounded, equicontinuous images under $S_d(\tau)$. A finite spatial net and finite grids for the image values on that net give finite covers in the uniform norm. Thus $S_d(\tau)$ maps bounded sets into relatively compact sets in X.

We will also use, for every C^2 Neumann-compatible g,

$$S_d(h)g - g = d\int_0^h S_d(r)\Delta g\,dr, \qquad \|S_d(h)g - g\|_\infty \le dh\|\Delta g\|_\infty. \tag{5}$$

For the identity, Green's identity gives

$$\langle \Delta g, \psi_j\rangle = -\lambda_j\langle g, \psi_j\rangle,$$

since both functions satisfy the Neumann condition. Integrating the multiplier $e^{-d\lambda_j r}$ proves (5) on each spectral coefficient, including the zero coefficient. The integral exists in X: contraction bounds every tail over an interval of length ε by $\varepsilon\|\Delta g\|_\infty$, so the truncated integrals are Cauchy in X. Completeness of the eigenbasis identifies the two sides in L^2; their continuity gives the identity in X. Contraction then gives the displayed norm estimate. These arguments use fixed positive heat times and are valid in every dimension.

Lemma 2.1. *Let $d > 0$, and let $u_t - d\Delta u = f$ be a classical solution with homogeneous Neumann conditions for positive times. Suppose that f is continuous in time with values in X, and that*

$$\sup_{t\ge 1}\|u(t)\|_\infty < \infty, \qquad F := \sup_{t\ge 1}\|f(t)\|_\infty < \infty.$$

Then $\{u(t) : t \ge 2\}$ has compact closure in X, and u is uniformly continuous from $[2,\infty)$ into X.

Proof. The forced heat formula is

$$u(t) = S_d(t-s)u(s) + \int_s^t S_d(t-r)f(r)\,dr, \qquad t \ge s \ge 1. \tag{6}$$

It follows by solving the scalar equation on each Neumann eigenfunction coefficient. The integrand is continuous in X away from its upper endpoint. The improper integral converges in X, because every endpoint tail of length ε has norm at most $F\varepsilon$. Completeness of the eigenbasis then gives the stated identity.

For $t \ge 2$ and $0 < \varepsilon < 1$, split the last unit time interval:

$$\begin{aligned} u(t) = S_d(1)u(t-1) &+ S_d(\varepsilon)\int_\varepsilon^1 S_d(r-\varepsilon)f(t-r)\,dr \\ &+ \int_0^\varepsilon S_d(r)f(t-r)\,dr. \end{aligned}$$

The first term ranges over a relatively compact set. The input of the middle heat operator is bounded by F, so the second term also ranges over a relatively compact set. The last term has

norm at most $F\varepsilon$. Taking ε small and covering the first two sets proves total boundedness of the orbit. Its closure is compact because X is complete.

For $\eta > 0$, choose a finite η-net with centers $g_j = u(t_j)$ from the actual orbit, $t_j \geq 2$. Each g_j is C^2 and Neumann-compatible. Contraction and (5), followed by (6), give

$$\|u(t+h) - u(t)\|_\infty \leq 2\eta + h\left(d \max_j \|\Delta g_j\|_\infty + F\right), \qquad t \geq 2.$$

The maximum is finite because there are finitely many centers. First choose η small, then h small. This proves uniform continuity; it requires no bound on $\Delta u(t)$ over the entire orbit. □

Lemma 2.2. *Let $d > 0$, and let w be a classical Neumann solution of $w_t - d\Delta w = Qf$, with $\bar{w}(t) = 0$. Suppose that $f(t) \in X$ is continuous, bounded for $t \geq 1$, and $\|f(t)\|_\infty \to 0$. Then $\|w(t)\|_\infty \to 0$.*

Proof. Set $\gamma = d\lambda_1 > 0$. The spectral expansion and (4) imply, for $r \geq 1$,

$$\|S_d(r)Qf\|_\infty \leq H_{d,1} e^{-\gamma(r-1)} \|Qf\|_2 \leq H_{d,1} e^{-\gamma(r-1)} \|f\|_\infty.$$

For $0 \leq r \leq 1$, contraction gives $2\|f\|_\infty$. Thus, for a finite constant C_d depending on d, Ω,

$$\|S_d(r)Qf\|_\infty \leq C_d e^{-\gamma r} \|f\|_\infty, \qquad r \geq 0. \tag{7}$$

Apply (6) from any fixed $T \geq 2$:

$$\|w(t)\|_\infty \leq C_d e^{-\gamma(t-T)} \|w(T)\|_\infty + C_d \int_T^t e^{-\gamma(t-s)} \|f(s)\|_\infty \, ds.$$

For a fixed $S \geq T$, the integral over $[T, S]$ tends to zero as $t \to \infty$, while the part over $[S, t]$ is at most $\gamma^{-1} \sup_{s \geq S} \|f(s)\|_\infty$. Taking $t \to \infty$ and then $S \to \infty$ proves the lemma. □

3 Proof of the attraction theorem

Proof of Theorem 1.1. Fix a uniform bound $K \geq 1$ for the absolute values of a, b, c. First, all three components remain nonnegative. To see this directly, write $U = (a, b, c)$, $p_i = U_i^-$, and $V = U^+$, and denote the reaction vector by $F(U) = (b(c-a), b(c-a), -b(c-a))$. On the set $p_i > 0$, the i-th reaction component satisfies $F_i(V) \geq 0$, whereas

$$|b(c-a) - V_b(V_c - V_a)| \leq 2K(p_a + p_b + p_c).$$

Testing the three equations by the negative parts, justified by a smooth convex approximation, gives

$$P_-(t) \leq 12K \int_0^t P_-(s) \, ds, \qquad P_-(t) := \int_\Omega (p_a^2 + p_b^2 + p_c^2) \, dx.$$

Here $P_-(0) = 0$. For $J(t) = \int_0^t P_-(s) \, ds$, $(e^{-12Kt} J(t))' \leq 0$, and $J \geq 0$; hence $P_- = 0$.

Integrating the equations preserves the two masses in (2). In particular, with $B(t) = \bar{b}(t)$,

$$\bar{a} = M_1 - M_2 + B, \qquad \bar{c} = M_2 - B, \qquad B' = \int_\Omega b(c-a) \, dx. \tag{8}$$

Since $c - a \geq -K$,

$$B(t) \geq B(0) e^{-Kt} > 0 \qquad \text{for every finite } t \geq 0.$$

Only this finite-time nonvanishing will be needed when normalizing by B.

Consider the quadratic energy of a, c:

$$E(t) = \frac{1}{2}\int_\Omega (a(t)^2 + c(t)^2)\,dx.$$

The Neumann conditions remove the boundary terms, and $(a-c)b(c-a) = -b(c-a)^2$ gives the exact dissipation identity

$$E' = -d_1\|\nabla a\|_2^2 - d_3\|\nabla c\|_2^2 - \int_\Omega b(c-a)^2\,dx \le 0. \tag{9}$$

Since $E \ge 0$, the two gradient squares are time-integrable. By (3),

$$z(t) := \|Qa(t)\|_2^2 + \|Qc(t)\|_2^2, \qquad \int_2^\infty z(t)\,dt < \infty.$$

The forcing $b(c-a)$ is uniformly bounded and continuous in X. Lemma 2.1 applies separately to a, c with their own diffusivities, and gives uniform continuity in time in X. The quadratic functional defining z is Lipschitz on bounded subsets of X; for example,

$$\big|\|Qg\|_2^2 - \|Qh\|_2^2\big| \le 8K\|g-h\|_\infty \quad \text{if } \|g\|_\infty, \|h\|_\infty \le K.$$

Thus z is uniformly continuous. A nonnegative uniformly continuous function with finite integral tends to zero: otherwise a separated sequence of times would give disjoint intervals of a common positive length on which the function stays bounded below. Therefore $z(t) \to 0$.

The same lemma makes the a, c orbits relatively compact in X. It follows that

$$h(t) := \|a(t) - \bar a(t)\|_\infty + \|c(t) - \bar c(t)\|_\infty \longrightarrow 0. \tag{10}$$

Indeed, a violating sequence has a common uniformly convergent subsequence for a, c. The limits have zero variance because $z \to 0$, and hence are constant, contradicting that violation.

It remains to identify these changing averages. Put $\delta = 2M_2 - M_1 > 0$. Equation (8) gives $\bar c - \bar a = \delta - 2B$, so the exact equation for B is

$$B' = B(\delta - 2B + \rho(t)), \qquad \rho(t) = \frac{1}{B(t)}\int_\Omega b\big[(c-\bar c) - (a-\bar a)\big]\,dx. \tag{11}$$

Because $b \ge 0$ and $\int_\Omega b = B > 0$,

$$|\rho(t)| \le h(t) \longrightarrow 0. \tag{12}$$

The weight b/B has total mass one, so this estimate remains valid even when B is small.

For $0 < \eta < \delta$, choose $T \ge 2$ with $|\rho(t)| \le \eta$ for every $t \ge T$. Then

$$B(\delta - \eta - 2B) \le B' \le B(\delta + \eta - 2B).$$

For $k_\pm = \delta \pm \eta$, define

$$y_\pm(t) = \frac{k_\pm/2}{1 + (k_\pm/(2B(T)) - 1)e^{-k_\pm(t-T)}}.$$

The denominators are positive, since $k_\pm > 0$ and $B(T) > 0$. These functions solve $y_\pm' = y_\pm(k_\pm - 2y_\pm)$ and start at $B(T)$. The difference $D = B - y_-$ satisfies

$$D' \ge (k_- - 2(B + y_-))D, \qquad D(T) = 0,$$

so multiplication by the corresponding positive integrating factor gives $B \ge y_-$. Applying the same calculation to $y_+ - B$ gives $B \le y_+$. Hence

$$\frac{\delta - \eta}{2} \le \liminf_{t\to\infty} B(t) \le \limsup_{t\to\infty} B(t) \le \frac{\delta + \eta}{2}.$$

Letting $\eta \downarrow 0$ proves

$$B(t) \longrightarrow M_2 - M_1/2, \qquad \bar{a}(t), \bar{c}(t) \longrightarrow M_1/2.$$

Together with (10), this identifies the uniform limits of a, c. In particular

$$\|b(t)(c(t) - a(t))\|_\infty \leq K\big(\|c(t) - M_1/2\|_\infty + \|a(t) - M_1/2\|_\infty\big) \to 0.$$

Finally $w = b - B$ has zero average and satisfies

$$w_t - d_2 \Delta w = Q\big(b(c - a)\big).$$

Lemma 2.2 gives $\|b(t) - B(t)\|_\infty \to 0$. The already determined limit of B proves the required limit of b, and completes the convergence of all three components. □

AI use

OpenAI Codex assisted with source retrieval, development and checking of the mathematical argument, and preparation of this manuscript. Two further evaluations in fresh Codex contexts provided reader and referee feedback on one frozen draft; they shared the same provider and workspace with the drafting process. The manuscript was then revised. Independent mathematical review is welcome.

References

[1] T. L. Nguyen and B. Q. Tang, *Stability analysis of irreversible chemical reaction-diffusion systems with boundary equilibria*, Zeitschrift für angewandte Mathematik und Physik **77** (2026), article 199, 30 pp. doi:10.1007/s00033-026-02847-0.

[2] F. Baudoin, *Diffusion processes and Stochastic calculus*, author version dated 13 February 2019, Chapter 4, Proposition 4.49 and Section 8, pp. 131, 135–136. https://fabricebaudoin.blog/wp-content/uploads/2019/12/book_emsstyle.pdf.